

- Q.27 Write about optimum thickness of insulation.
- Q.28 Explore Newton's law of cooling and its application in determining heat transfer rates in convective system.
- Q.29 Explain the concept of Log Mean Temperature Difference (LMTD) and its significance in heat exchanger analysis.
- Q.30 Discuss Kirchoff's law and its implications for radiative equilibrium.
- Q.31 Differentiate between natural and forced circulation evaporators
- Q.32 Explain the phenomena of entrainment and foam formation in evaporators and their effects on performance.
- Q.33 Explain about finned tube heat exchangers.
- Q.34 Explain about the Wein's displacement law, Stefan - Boltzman Law.
- Q.35 Write about the types of condensation with example.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Describe the construction and working principles shell and tube heat exchangers
- Q.37 Illustrating the application of various feeding methods and multiple effect evaporation in industrial processes.
- Q.38 Write short note on any three:-
- condensation and
 - Grashoff No
 - Scale formulation
 - Factors influencing the selection of insulation materials

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Roll No.

4th Sem / P & P, Chem Engg. (Spl. Paint Tech), Chem Engg. (Spl. Polymer Engg.)

Subject:- Heat Transfer

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What property determines a material's ability to conduct heat?
- Thermal diffusivity
 - Density
 - Specific heat
 - Thermal conductivity
- Q.2 Radiation heat transfer does not require:
- Medium for propagation
 - Temperature difference
 - Surface roughness
 - Direct contact between bodies
- Q.3 Which types of heat exchanges offers the highest heat transfer surface area per unit volume?
- Double pipe heat exchanger
 - Plate heat exchanger
 - Shell and tube heat exchanger
 - Regenerative heat exchanger
- Q.4 Multiple effect evaporation is used to :
- Increase the efficiency of evaporators
 - Reduce the size of evaporators
 - Enhance the quality of evaporated products
 - Minimize energy consumption

- Q.5 The primary purpose of insulation in heat transfer system is to:
- Increase heat transfer rate
 - Reduce thermal conductivity
 - Eliminate temperature gradients
 - Minimize heat loss
- Q.6 Which property does not affect the rate of convective heat transfer?
- Fluid velocity
 - Fluid viscosity
 - Surface roughness
 - Material conductivity
- Q.7 The effectiveness of a heat exchanger is defined as the ratio of:
- Actual heat transfer to the maximum possible heat transfer
 - Overall heat transfer coefficient to the surface area
 - Heat transfer rate to the temperature difference
 - Mass flow rate to the heat capacity
- Q.8 What is the name of the dimensionless number used to characterize the relative importance of conduction and convection in a fluid flow?
- Prandtl number
 - Reynolds number
 - Nusselt number
 - Peclet number
- Q.9 Critical thickness of insulation refers to the:
- Maximum thickness for effective insulation
 - Minimum thickness for safety
 - Thickness at which conduction dominates over convection
 - Thickness at which radiation becomes significant

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- Q.10 Entrainment in evaporators refers to the:
- Formation of foam
 - Leakage of steam
 - Carryover of liquid droplets with vapor
 - Accumulation of scale

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Write formula of Reynold number?
- Q.12 Define black body?
- Q.13 Write Fourier's law?
- Q.14 What is conduction
- Q.15 Write name a common conductor material.
- Q.16 Define kirchoff law.
- Q.17 Name one application of heat transfer,
- Q.18 Full form of LMTD
- Q.19 State Insulation
- Q.20 Name one example of convection mode of heat transfer.

SECTION-C

- Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)
- Q.21 Explain the concept of thermal conductivity and discuss how it varies for different materials.
- Q.22 Discuss about emissivity in radiation mode of heat transfer and write its practical application.
- Q.23 Describe the analysis of heat transfer through multilayered flat walls.
- Q.24 Difference between parallel flow and countercurrent flow with neat and clean diagram.
- Q.25 Discuss about Double pipe heat exchangers with neat and clean diagram.
- Q.26 Difference between steady-state and unsteady -state heat transfer.

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